TRINOMIAL CUBIC SUMS: AN AUDIT OF EXISTING PROOFS AND ANNOTATED REVIEW

DEEPMATH06 - NUMBER THEORY

Abstract. We audit the existing trinomial cubic-sum research in the deepmath06 export, separating complete algebraic certificates, local textual errors, substantive proof gaps, finite evidence, and false formulas. The third-order recurrence, Newton ratio envelope, archived low-order polynomial criteria, and a negative-square moment certificate pass verification. Several higher-order classifications and general parameter assertions still lack effective tail arguments. Every annotation identifies its original asset and line range. The original research files remain unchanged. This is an audit of supplied material, not a research paper presenting repairs or promoting unfinished assertions to theorems.

Contents

1. Scope, definitions, and verdicts 1
2. Overview and evidence boundary 2
3. Assertion-by-assertion ledger 3
3.1. Basic representations, recurrence, and factorization 3
3.2. Ratio bounds, monotonicity, and logarithmic differences 4
3.3. Root differences and cancellation 6
3.4. Arbitrary order, common tails, and moment theory 6
3.5. General color counts, powers, and weights 7
4. Annotations of errors, gaps, and insufficient rigor 9
5. Exact checks of archived certificates 15
5.1. The third-order recurrence 15
5.2. The Newton ratio envelope 15
5.3. The archived criteria for S_2 and S_3 16
5.4. Weighted identity and local corrections 16
5.5. Negative-square and finite counterexamples 17
6. Coverage and reproducibility 17
6.1. Computational ranges 18
6.2. Evidence files and integrity of the inputs 18
7. Verification record and scope 18
References 20

1. Scope, definitions, and verdicts

The audit date is 10 November 2026. The inputs are the offline export, especially its 21 current research assets A000001–A000021 and associated notes, scripts, and data. We check existing assertions, existing proof routes, and finite ranges already

claimed in the sources. We do not search for new recurrences, scan new parameter regions, investigate novelty, or edit the original proofs. Exact recomputations that check an archived induction certificate or polynomial criterion are stored separately with this report.

Throughout,

$$a_n = \sum_{i+j+k=n} \binom{n}{i,j,k}^3, \qquad r_n = \frac{a_n}{a_{n-1}}, \qquad b_n = \frac{a_{n+1}^{1/(n+1)}}{a_n^{1/n}}, \tag{1.1}$$

$$f_n = \log a_n, \quad g_n = \frac{f_n}{n} \quad (n \geq 1), \tag{1.2}$$

$$S_k(n) = (-1)^k \Delta^k f_n, \qquad U_k(n) = (-1)^{k+1} \Delta^k g_n. \tag{1.3}$$

Here $\Delta h_n = h_{n+1} - h_n$, $a_0 = 1$, and $f_0 = 0$. If the definition is extended to order zero, it gives $U_0 = -g$.

P : A complete certificate, or all necessary algebraic conditions in an archived proof route, has been verified and supports the specified conclusion.

L : The conclusion is verifiable, but equality cases, signs, summation steps, or similar local details in the source need correction.

G : A substantive step or an unestablished hypothesis remains. This is not a verdict that the proposition is false or that it is open in the literature.

N : Only finite computational evidence is confirmed. Unchecked indices are not silently included in a proof.

F : A specific assertion, formula, or designated certificate is wrong. The incorrect object must be distinguished from the theorem it was intended to prove.

O : The source explicitly leaves the question open, or the archive does not complete it.

One source can contain assertions with different verdicts. The ledger counts assertion units, not independent new theorems. Successful typesetting is not evidence of mathematical correctness.

2. Overview and evidence boundary

The original symmetric-cube script returns zero remainder. The two archived data tables contain $a_0, \ldots, a_{100}$ and $a_0, \ldots, a_{200}$; both match an independent evaluation from the definition. The Newton ratio certificate and archived S_2, S_3 criteria pass exact algebraic checks. The four original conclusions can be verified through these archived routes, although the invalid Perron presentations must remain annotated. Verification of strict decrease of b_n uses the weighted identity, whose printed proof requires local corrections.

By contrast, the all-index classifications of S_4, S_5, S_6, the derived complete tail classifications of U_3, U_4, U_5, the special cubic common-tail assertion, and the proposed general parameter thresholds cannot be certified from finite scans and unproved remainder percentages. Exact cross-checks provide explicit counterexamples to three universal or literal assertions: the all-order initial negative-sign rule, the common-tail theorem for every parameter pair, and the displayed simplex moment formula. These are recorded as false assertions rather than merely missing proofs. Concrete negative values, finite cancellation checks, and $L(p^2) = -206$ remain rigorous finite certificates.

Code	Verdict	Units
P	Complete certificate / verified existing route	20
L	Verified conclusion; local corrections to the text	4
G	Substantive proof gap or unestablished hypothesis	14
N	Finite computational evidence	2
F	An explicit assertion, formula, or certificate is wrong	10
O	Explicitly open or unfinished in the supplied archive	3

3. Assertion-by-assertion ledger

3.1. Basic representations, recurrence, and factorization.

C01 [P] Franel convolution, generating function, and probability representation

These identities follow from rearranging finite sums or multiplying formal series. They agree with the independent evaluation of $a_0, \dots, a_{200}$ from the defining sum. For the generating function, also see [6], lines 5–8.
Source: [1], lines 1–28.

C02 [P] $\theta^3\Phi = z\Phi$ and finite closure for integer powers

Termwise differentiation verifies the basic equation. For integer powers, finitely many derivative monomials give a finite-dimensional closure. This proves existence of some recurrence, not the minimal order of a guessed recurrence.
Source: [6], lines 3–14.

C03 [L] $r_n < 27$ for every $n \geq 1$

The power-mean/probability route is valid after correcting its reversed equality condition; see F01. The verified tail envelope and existing finite initial segment also support the conclusion.
Source: [3], lines 1–39.

C04 [G] The supplied proof of $a_n \sim \frac{9}{4\pi^2} 27^n n^{-2}$

The local Gaussian calculation and its constant are consistent, but the text supplies only a proof sketch. It does not establish a uniform central approximation and a negligible complementary tail. This sketch is not counted as a complete proof of the full asymptotic statement.
Source: [3], lines 34–55.

C05 [P] $r_n \to 27$ and $b_n \to 1$

The archived Newton envelope has passed exact verification, and both bounds tend to 27. The ratio limit gives the root limit by averaging logarithms, hence the stated limit of b_n. The separate Perron argument in A000009 is invalid; see F02.
Source: [15], lines 13–85.

C06 [P] The displayed third-order polynomial recurrence

Execution of the original symmetric-cube verification script leaves no nonzero term. Coefficient extraction therefore proves the recurrence for all indices. Independently, the 201 terms evaluated from the definition give zero residual for $n = 0, \dots, 197$.
Source: [6], lines 16–96.

C07 [P] $-P_0, P_1, -P_2, P_3 > 0$ and the limiting characteristic polynomial
The coefficient signs verify these inequalities, and the limiting polynomial is $(\lambda - 27)(\lambda^2 + 27)$. This does not make all three terms on the solved recurrence right-hand side positive; see F03.
Source: [4], lines 14–25.

C08 [P] The first five formal coefficients and their logarithmic transform
The listed $c_1/c_0, \ldots, c_5/c_0$ pass the relevant formal recurrence coefficient checks. Their stated transforms $d_1, \ldots, d_5$ agree. This verifies formal algebra only, not existence of an asymptotic expansion for the actual solution or effective remainder bounds.
Source: [4], lines 29–41.

C09 [F] The displayed n^{-6} coefficient of $r_n/27$
The value $-178/2187$ is inconsistent with the first five coefficients printed in the same asset. Taking their formal quotient gives $82/729$ at order six; see F05.
Source: [4], lines 45–51.

C10 [P] $P_1 = 3(3n + 5)Q_5$ and the quadratic discriminants
Expansion verifies the factorization. Both quadratic discriminants equal -35, and the difference of the two quadratics is $14(3n + 5)$.
Source: [5], lines 1–12, 26–30.

C11 [P] Irreducibility of Q_5 over $\mathbb{Q}$
The specified irreducibility test modulo 19 passes exact verification. Primitivity and the nonvanishing leading coefficient make this a valid certificate of irreducibility over the rationals.
Source: [5], lines 3–12.

C12 [G] The supplied irreducibility proof for R_6 over $\mathbb{Q}$
The written certificate modulo 37 fails: reduction modulo 37 has a quadratic factor. This is a failed proof, not a claim that R_6 is reducible over $\mathbb{Q}$; see F06.
Source: [5], lines 14–26.

3.2. **Ratio bounds, monotonicity, and logarithmic differences. C13 [P] $L_n < r_n < U_n$ for $n \geq 20$, with $C = 20$**
The initial cases, mixed-corner argument, and Newton certificates for the two degree-twelve numerators pass exact verification. The top differences are 4345502515200 and 278112160972800, as reported.
Source: [15], lines 9–85.

C14 [P] Strict increase of r_n, or $S_2(n) > 0$ for all indices
The original Perron-plus-finite-check presentation is incomplete. A separate archived envelope criterion, $L_{n+2} > U_{n+1}$, does pass polynomial positivity verification. The criterion occurs in [24], `rigorous_tail_bounds2.py`, lines 116–118. Together with the existing finite segment, this existing route closes the assertion; see the certificate section.
Source: [16], lines 37–41.

C15 [L] Strict decrease of b_n

The original asymptotic-plus-finite-check presentation has a gap. The archive also contains a route through $S_3 > 0$ and the weighted root-difference identity, which gives $U_2 = -\Delta^2 g > 0$. The identity proof needs the local textual corrections in F10.
Source: [4], lines 77–93.

C16 [P] $S_3(n) > 0$ for all indices

The archived H_3 recurrence criterion passes exact verification: its derivative signs and lower corner bound reduce to polynomials with positive coefficients after $n = 20 + t$. The finite segment passes integer comparisons.
Source: [16], lines 43–53.

C17 [G] $S_4(n) > 0$ for all indices

The finite evidence passes. No complete effective tail certificate is established in the finished texts. Some candidate scripts assume a finer ratio envelope as a proven handoff without supplying its induction certificate.
Source: [16], lines 169–185.

C18 [G] The complete single-exception classification of S_5

The inequality $S_5(3) < 0$ is rigorously confirmed. Positivity at every other index still depends on an unestablished high-order tail remainder bound.
Source: [15], lines 97–150.

C19 [G] The complete two-exception classification of S_6

The inequalities $S_6(0), S_6(3) < 0$ are rigorously confirmed. Positivity at every other index does not follow from the remainder arguments supplied.
Source: [15], lines 103–150.

C20 [P] $S_k(n) - S_k(n+1) = S_{k+1}(n)$

This follows by expanding the forward difference. The conditional deduction of lower-order tail positivity from higher-order positivity and convergence to zero is also correct.
Source: [10], lines 123–141.

C21 [G] The stated thresholds 21 and 500, and percentage remainder bounds

A positivity threshold for the first two terms of a truncated expansion does not control the remaining terms. A000014 also has inconsistent formula and table values for its sixth-order threshold; see F07 and F08.
Source: [12], lines 57–95.

C22 [P] $S_5(3) < 0$, $S_6(0) < 0$, and $S_6(3) < 0$

All three signs are confirmed by exact integer product comparisons using the correct binomial exponents. No floating-point logarithm is needed for these finite counterexamples.
Source: [10], lines 153–155.

C23 [N] The high-precision scan through 5000

A 120-digit Decimal repetition within the range already claimed in the archive gives the same sign pattern. Independent integer verification covers the ranges supported by $a_0, \ldots, a_{200}$; the larger scan remains numerical evidence.
Source: [16], lines 55–66.

3.3. **Root differences and cancellation. C24 [L] The weighted identity for root differences**

The principal identity is correct, with $U_0 = -g$ under the stated definition. Its integrating-factor summation has incorrect intermediate expressions and an incorrect zero-order sign check. The correct telescoping weight verifies the principal identity from the archived recurrence; see F10.
Source: [11], lines 17–70.

C25 [L] $U_1, U_2 > 0$ **for all indices**

These follow from the verified positivity of S_2, S_3 and the positive weighted sum in the principal identity. The local textual errors in that identity proof must remain marked.
Source: [21], lines 33–37.

C26 [G] $U_3 > 0$ **for all indices**

The written proof depends on all-index positivity of S_4 and inherits the substantive gap in C17.
Source: [21], lines 33–37.

C27 [G] $U_4 > 0$ **for all indices**

The finite cancellation $U_4(4) > 0$ is rigorously confirmed. The global argument still requires $S_5(j) > 0$ for every $j \geq 4$, and therefore inherits the gap in C18.
Source: [11], lines 97–131.

C28 [P] $U_5(1) < 0$ **and the finite cancellation checks**

Clearing rational denominators and comparing integer powers confirms $U_5(1) < 0$ and $U_5(2), U_5(3), U_5(4) > 0$. These finite conclusions are recorded separately from the full tail classification.
Source: [11], lines 135–156.

C29 [G] $U_5(n) > 0$ **for every** $n \geq 2$

The tail proof assumes $S_6(j) > 0$ for every $j \geq 4$. A000021 does not fill this gap and, in one sentence, also overlooks the negative term at $j = 3$.
Source: [11], lines 145–161.

C30 [P] The conditional finite-negative-term cancellation principle

If the negative index set is known to be finite, all other terms are strictly positive, and the necessary partial sums are checked, then the numerator increases after its final negative increment. This conditional argument does not establish its hypotheses for a particular sequence.
Source: [11], lines 165–173.

C31 [F] The two asymptotic formulas for $U_k(n)$

The first contains an extra factor $k!$; the purportedly more precise second formula omits the leading factor $\log n$. They cannot both describe the same quantity; see F11.
Source: [11], lines 205–216.

3.4. **Arbitrary order, common tails, and moment theory. C32 [G] The supplied proof of eventual positivity at every fixed order**

A sufficiently high-order expansion of the actual solution would support the expected deduction. The present text instead differences a low-order pointwise

remainder to arbitrary order, without establishing the asserted remainder order or explicit constants.
Source: [17], lines 9–77.

C33 [G] Nonexistence of a common positive tail for all orders
The finite tables and negative signs at $n = 0$ do not imply that negative indices are unbounded as the order varies. The required quantifier step is missing.
Source: [17], lines 81–148.

C34 [F] $S_k(0) < 0$ **for every** $k \geq 6$
This universal assertion is false. Exact rational logarithm intervals certify $S_{21}(0) > 0$ using initial terms already contained in the archived data table. The original $6 \leq k \leq 20$ negative signs remain finite evidence, not a general rule; see F29.
Source: [17], lines 97–117.

C35 [P] $L(p^2) = -206$ **and nonexistence of a positive real moment measure**
For $p = (x-1)(x-11)(x-20)$, the square expansion and exact substitutions are correct. The negative square excludes a positive moment measure anywhere on the real line.
Source: [17], lines 152–189.

C36 [P] $\det H_4 < 0$
The exact value is -49203288, confirming this finite certificate. The signs of the tabulated determinants $H_1, \ldots, H_7$ also agree.
Source: [17], lines 191–205.

C37 [F] Negative definiteness of H_n **for every** $n \geq 4$
A negative determinant is not negative definiteness. In fact, $e_0^T H_n e_0 = a_0 = 1 > 0$ immediately rules out negative definiteness.
Source: [17], lines 191–205.

C38 [O] Positive moment representations of shifted tails
The source reports a finite random search and explicitly states that it is not a proof. This audit leaves the question open and conducts no new search.
Source: [17], lines 207–223.

3.5. General color counts, powers, and weights. C39 [P] P-recursive existence for integer color counts and integer powers

The basic differential equation and finite closure of derivative monomials give some ODE. Coefficient extraction and the factorial rescaling then give some polynomial recurrence. This argument concerns integer parameters.
Source: [7], lines 10–16.

C40 [G] The supplied general growth and ratio-limit proof
The source acknowledges unfinished rigorous error analysis, and its power of n contradicts the special $(3, 3)$ formula in the same archive. The ratio limit itself is not refuted by this incorrect prefactor.
Source: [7], lines 18–30.

C41 [F] The prefactor $n^{-(k-1)m/2}$ in the general asymptotic formula

The already studied case $(k, m) = (3, 3)$ gives n^{-3}, contradicting the specialized n^{-2} formula. The multinomial identity at $m = 1$ also shows that this prefactor cannot hold generally.
Source: [13], lines 149–161.

C42 [O] Minimal recurrence order $\min(k, m)$ and the claimed general $m = 2$ cases

The general minimal-order statement remains conjectural. Exhibiting a recurrence does not prove minimality, and the symmetric-power module dimension is not the coefficient recurrence order. Several paragraphs labeled proved provide insufficient justification.
Source: [13], lines 174–197.

C43 [N] Exact classifications of $K(2), K(3), K(4), K(5)$

An exhibited negative value gives the relevant upper bound. The lower bound requires all-index positivity and cannot follow from a scan to 30 plus an unspecified eventual threshold. For $d = 3$, the S_4 proof gap remains.
Source: [18], lines 52–88.

C44 [O] General $K(d, s)$ classification, real color parameters, and joint growth

These portions remain conjectures, untested candidates, or frameworks without rigorous errors in the source. Project-level completion labels do not replace proofs.
Source: [18], lines 73–128.

C45 [F] $A_n^{(d)} \sim d^n$ for fixed n and $d \to \infty$

The already defined cubic sum at $n = 2$ is $A_2^{(d)} = d + 8\binom{d}{2} = 4d^2 - 3d$. The stated asymptotic equivalence omits a constant factor.
Source: [18], lines 92–104.

C46 [G] Fixed-order asymptotics for general weights and real powers

The proposed leading coefficient has a plausible origin, but the text does not establish a sufficiently high-order uniform expansion with controllable differences. The ordinary finite-order D-module argument also does not apply directly to non-integer powers.
Source: [19], lines 19–43.

C47 [P] The degeneration at $s = 1$

Probability normalization gives $Z_n(1) = 1$. In the unweighted model, $A_n^{(d,1)} = d^n$, so every logarithmic difference of order $k \geq 2$ vanishes.
Source: [19], lines 39–43.

C48 [P] The fifth-power binomial and weight-30 counterexamples

Termwise evaluation confirms $B_1B_3 - B_2^2 = -180$ and $W_0W_2^3 - W_1^3W_3 = -66992580$. These are checks of existing concrete counterexamples.
Source: [19], lines 47–112.

C49 [F] The displayed $s = 2$ simplex moment formula

The literal displayed integral is false: the already included uniform $d = 2$ case gives moments $1, 1/2, 3/8$, incompatible with the proposed simplex kernel, whose minimum is $1/2$. This does not refute other possible moment representations. The separate unfinished normalization remains an additional gap; see F28.

Source: [19], lines 124–142.

C50 [F] The proposed inequality $K(d,5) > K(d,3)$ for all d

The archived $d = 2, s = 5$ counterexample gives $K(2,5) = 1$. Since the definition always gives $K \geq 1$, this strict inequality cannot hold for every d.
Source: [20], lines 105–109.

C51 [G] The uniform explicit threshold $N(d,s,k)$

The parameter scaling is not derived from a next-term coefficient and a remainder bound. The derivation omits the asserted parameter factors, and the limiting process in $o_{d,s}(1)$ is unspecified.
Source: [20], lines 113–139.

C52 [F] Nonexistence of a common tail for general parameters

This universal assertion is false. In the already included case $(d,s) = (2,2)$, the central binomial sequence has $S_k(n) > 0$ for every $k \geq 2$ and $n \geq 0$, so $N = 0$ is a common positive tail. See F27. The special $(3,3)$ assertion C33 remains unproved in the supplied material.
Source: [20], lines 143–153.

C53 [F] The final synthesis: moment and complete-monotonicity conditions

Positive moment measures require positive semidefinite Hankel matrices and nonnegative determinants, not strict positivity of every determinant. The text also interchanges differences of the sequence and of its logarithm; see F21.
Source: [20], lines 155–167.

4. Annotations of errors, gaps, and insufficient rigor

Each annotation concerns a specific original passage. It records severity, location, and scope of impact without editing the source proof.

Audit annotation 4.1 (F01: Reversed equality condition in the power-mean step). **Local error.** Source: [3], lines 15–26.

The source states that equality in $(x+y+z)^3 \leq 9(x^3+y^3+z^3)$ requires at most one nonzero entry. Equality actually requires $x = y = z$; $(1,0,0)$ already refutes the printed condition. Strictness can be checked at a boundary term of the probability recurrence. This needs a correction, but it does not make $r_n < 27$ false.

Audit annotation 4.2 (F02: The Perron exceptional subspace is not excluded as written). **Major proof error.** Source: [9], lines 33–38.

The lower bound $\liminf a_n^{1/n} \geq 3$ does not exclude solutions with growth rate at most $\sqrt{27}$, since $3 < \sqrt{27}$. Moreover $r_1 = 3$, not strictly greater than 3. The audit confirms the ratio limit using the separate archived Newton envelope, not this invalid Perron argument.

Audit annotation 4.3 (F03: An old note reverses the middle recurrence sign). **Formula error.** Source: [22], lines 22–26.

Solving the original recurrence gives $a_{n+3} = (-P_0 a_n - P_1 a_{n+1} - P_2 a_{n+2})/P_3$. The old note prints $+P_1 a_{n+1}$ and calls the recurrence positive. The later verification script retains the correct sign, so this old error does not invalidate C06.

Audit annotation 4.4 (F04: Formal series, pointwise errors, and difference errors are conflated). **Substantive gap.** Source: [17], lines 21–65.

A bound $|E_n| \le Kn^{-4}$ alone gives only a crude $O(n^{-4})$ bound for a fixed-order difference; it does not automatically improve to $O(n^{-k-1})$. Formal coefficient matching does not by itself establish a full expansion for the actual solution. A sufficiently high-order established expansion would permit an appropriate truncation before differencing, but that premise is not supplied. A000010, lines 73–79, and A000017, lines 49–55, also have sign slips: the leading sign of $\Delta^k \log n$ is $(-1)^{k-1}$, and that of $\Delta^k(-2\log n)$ is $(-1)^k$.

Audit annotation 4.5 (F05: The sixth ratio coefficient contradicts the preceding coefficients). **Coefficient error.** Source: [4], lines 35–51.

Put $h(t) = 1 + \sum_{j=1}^{5}(c_j/c_0)t^j$. For the rational coefficients printed in the same asset, $(1-t)^2 h(t)/h(t/(1-t))$ has coefficient $82/729$ at order six, not $-178/2187$. The omitted c_6 cancels at this order of the quotient, so omission of higher coefficients does not explain the inconsistency.

Audit annotation 4.6 (F06: The claimed irreducibility modulo 37 is false). **Failed certificate.** Source: [5], lines 14–26.

Exact reduction gives

$$R_6(n) \equiv 4(n^2 - 13n + 1)(n^4 + 13n^3 - 15n^2 - 8n - 12) \pmod{37}.$$

Thus the claimed exhaustive exclusion of quadratic factors is wrong. A rational-factorization routine returning no factor does not validate this particular modular certificate. This audit does not search for a replacement prime.

Audit annotation 4.7 (F07: The claimed tail remainder percentages are unproved). **Substantive gap.** Source: [16], lines 122–147.

A threshold at which the leading term exceeds the next term does not control the remaining terms. A000016 asserts a total relative error below 5% for every $n \ge 500$ without establishing the required estimate. A000015 also treats the leading value at $n = 500, k = 3$ as a lower bound for all larger indices and orders; $240/500^6$ is already much smaller, and these leading terms tend to zero as n grows.

Audit annotation 4.8 (F08: The log-ratio expansion and sixth-order threshold are inconsistent). **Formula and arithmetic errors.** Source: [15], lines 130–148.

Taking the logarithm of the displayed ratio expansion gives $\log r_n = \log 27 - 2/n - 2/(9n^2) + 2/(9n^3) + \cdots$, not the expression obtained by copying the ratio coefficients. Separately, A000014, line 51, gives $\max(20, \lfloor 6(9\cdot 6+7)/18 \rfloor + 3) = 23$, whereas its line-62 table says 21.

Audit annotation 4.9 (F09: The fifth-difference exponents were copied incorrectly). **Integer-criterion error.** Source: [11], lines 116–124.

The correct exponent vector is $(1, -5, 10, -10, 5, -1)$. Several fractions in A000011 and A000014 replace the fifth power of a_{n+1} by its first power and place a_n on the wrong side. They are not equivalent to the sign of S_5. A000012, line 16, has the correct criterion; the audit uses the defining exponents.

Audit annotation 4.10 (F10: The weighted identity is correct, but intermediate and zero-order checks fail). **Local proof errors.** Source: [11], lines 41–70.

The archived recurrence $(n+k+1)H_{n+1} = nH_n + \Delta^{k+1} f_n$ is correct. Its telescoping quantity is $n\binom{n+k}{k}H_n$. The sum in line 56 is not its solution, and lines 60–64 are inconsistent. By definition $U_0 = -g$; lines 68–70 instead write $U_0 = g$ and identify $-f_n/n$ with f_n/n. The principal identity remains verifiable directly from the archived recurrence after these local corrections.

Audit annotation 4.11 (F11: An extra factorial and a missing logarithmic leading factor). **Asymptotic formula errors.** Source: [11], lines 205–216.

In the source line-209 computation, the sum is asymptotic to $2\log n$, and the denominator to $n^{k+1}/k!$. If its premises hold, it therefore gives $2k!\log n/n^{k+1}$. Line 213 has an extra $k!$, while line 216 omits $\log n$ entirely. These cannot be an asymptotic equivalent and a more precise expansion of the same expression.

Audit annotation 4.12 (F12: The stage-three review does not independently complete stage two). **Dependency gap.** Source: [21], lines 33–88.

The all-index conclusions for U_3, U_4, and $U_5 (n \geq 2)$ depend on the all-index signs of S_4, S_5, S_6. A valid finite cancellation does not prove every later increment positive. A000021, line 88, also describes $S_6(j)$ for $j \geq 1$ as positive, overlooking the archived negative term at $j = 3$.

Audit annotation 4.13 (F13: Initial-index signs and finite tables do not exclude a common tail). **Quantifier gap.** Source: [17], lines 95–143.

The required assertion is that for every N some $n \geq N$ and some order have a negative value. Negativity fixed at $n = 0$ does not meet this quantifier. A table of finitely many orders cannot show that the largest negative index is unbounded; lines 123 and 141 promote the observed pattern to precisely that missing assertion.

Audit annotation 4.14 (F14: A negative determinant is called negative definiteness). **Conceptual error.** Source: [17], lines 191–205.

The negative square and negative fourth determinant already disprove a positive moment representation. However, every Hankel matrix here has upper-left entry 1 and hence cannot be negative definite. The tabulated signs for $4 \leq n \leq 7$ also do not prove that every larger determinant remains negative.

Audit annotation 4.15 (F15: The general power of n contradicts an archived special case). **Formula error.** Source: [13], lines 149–161.

The proposed factor $n^{-(k-1)m/2}$ gives n^{-3} at $(k, m) = (3, 3)$, contrary to the dedicated n^{-2} calculation. It cannot serve as the general theorem claimed. A000018 has a separate fixed-n, large-color-count error, already refuted at $n = 2$ by $A_2^{(d)} = 4d^2 - 3d$.

Audit annotation 4.16 (F16: Recurrence order, differential order, and module dimension are mixed). **Unproved generalization.** Source: [13], lines 174–197.

The text claims minimal coefficient recurrence order two for arbitrary $m = 2$ and arbitrary $k = 2$, without proving the required general statements. Differential order and coefficient recurrence order are distinct quantities. A000007 itself lists the general order as open, but A000020, line 31, uses the conjecture as an established premise.

Audit annotation 4.17 (F17: The lower bounds for the all-index positivity order are unproved). **Promotion of finite evidence.** Source: [18], lines 52–88.

The definition of $K(d)$ quantifies over every $n \geq 0$. A concrete negative value supplies a failure-order upper bound. A scan through 30 supplies no all-index lower bound without an effective certificate for every remaining index. No such certificate is completed in the general-parameter material.

Audit annotation 4.18 (F18: Noninteger powers are outside the ordinary finite-order D-module argument). **Scope error.** Source: [20], lines 19–43.

The ordinary differential operator θ^s used here has the required finite-order meaning only for integer s. The integer-order differential equation and symmetric-power argument cannot simply be applied to noninteger s. Even a correct leading term needs a sufficiently high-order expansion with controlled differences.

Audit annotation 4.19 (F19: The fifth-power positivity order cannot always be larger). **Contradicted by an archived counterexample.** Source: [20], lines 105–109.

A000020, line 108, proposes $K(d,5) > K(d,3)$ for every d. The stage-six exact second-order counterexample gives $K(2,5) = 1$, while the definition gives $K \geq 1$ in every case. The strict inequality therefore cannot be universal. The separate nondecreasing conjecture also conflicts with the same text's asserted value $K(2,3) = 3$.

Audit annotation 4.20 (F20: The uniform threshold lacks parameter-dependent error control). **Unproved quantitative assertion.** Source: [20], lines 113–139.

Line 119 contains $(s-1)^{-1}(d-1)^{-1}$, but the line-131 derivation has neither factor. An increased leading amplitude does not itself control parameter changes in correction terms. The limiting process in $o_{d,s}(1)$ is unspecified. This is not an effective uniform tail certificate.

Audit annotation 4.21 (F21: Positive moments, complete monotonicity, and logarithmic differences are confused). **Incorrect necessary conditions.** Source: [20], lines 155–167.

Hankel matrices of positive moment sequences must be positive semidefinite, so their determinants may vanish. A single point mass refutes strict positivity of every determinant. Line 165 refers to logarithmic differences but writes differences of the original sequence; line 167 then uses S_6 of $\log A$ against a condition on A. The object of the argument has changed.

Audit annotation 4.22 (F22: Several scripts labeled rigorous are not certificates). **Implementation defects.** Source: [23], lines 214–364.

The polynomial-inequality script calls parameter-requiring A_poly, B_poly, and D_poly without arguments. Its Sturm remainder uses integer floor division rather than rational polynomial division, and it calls the bounded interval ending at 10^8 an interval to infinity. Other scripts infer positivity at every later point from one sampled positive value, or trust a handoff hypothesis without a certificate. These outputs do not prove a complete tail.

Audit annotation 4.23 (F23: Finite cancellation, size estimates, and reused notation). **Local indexing and notation errors.** Source: [11], lines 175–241.

At $n = 4$ the weighted sum ends at $j = 3$, but line 180 invokes $S_5(4)$, which is not yet included. Line 241 says $a_0, \dots, a_8$ are below 10^6, although $a_8 = 912018123$. A000004, line 39, also reuses b_n for $a_n/27^n$, conflicting with the root-ratio definition.

Audit annotation 4.24 (F24: Historical assets and completion labels are not proof verdicts). **Evidence boundary.** Source: [2], lines 32–65.

A000001, A000002, and A000008 retain historical statements that the recurrence or explicit verification is unfinished. The later A000006 certificate resolves this particular issue. Conversely, the OEIS absence and novelty claims lack a preserved checkable search record; this audit conducts no external novelty investigation. Titles such as final version or independent review are not acceptance criteria.

Audit annotation 4.25 (F25: A missing leading constant and an extra root-ratio factor). **Formula errors.** Source: [4], lines 39–41, 83.

The normalization $a_n/27^n$ in line 41 omits the nonunit leading constant c_0 while using asymptotic equivalence. The root-ratio identity in line 83 inserts an extra factor 27: the powers of 27 cancel in $a_{n+1}^{1/(n+1)}/a_n^{1/n}$. These errors do not invalidate the separate verified monotonicity route.

Audit annotation 4.26 (F26: The printed remainder envelopes contradict known coefficients). **False quantitative bounds.** Source: [9], lines 93–96.

The proposed bound around $27(1 - 2/n + 1/n^2)$ cannot have an $O(n^{-3})$ error: its second coefficient differs from the established $16/9$ by $7/9$. At $n = 20$ the scaled error already exceeds the claimed constant 100. Separately, A000004, line 69, claims $C \le 1$ in a lower bound whose own difference expansion has next coefficient $-50/9$; its exact $n = 20$ check also fails.

Audit annotation 4.27 (F27: An archived central-binomial case has a common positive tail). **False universal theorem.** Source: [20], lines 145–153.

The source claims nonexistence of a common tail for every $d \ge 2, s > 1$. Its already included case $(d, s) = (2, 2)$ is $A_n = \binom{2n}{n}$. Put

$$h_n = \log(n+1) - \log(n+1/2) = \int_0^\infty \frac{e^{-(n+1/2)t} - e^{-(n+1)t}}{t}\, dt.$$

Since $\Delta \log A_n = \log 4 - h_n$, for every $k \ge 2$ and $n \ge 0$ one has

$$S_k(n) = \int_0^\infty (1 - e^{-t})^{k-1} \frac{e^{-(n+1/2)t} - e^{-(n+1)t}}{t}\, dt > 0.$$

The integrals converge at both endpoints. Thus $N = 0$ is a common tail. This checks an existing universal assertion against a case already in the archive; it is not a new parameter classification.

Audit annotation 4.28 (F28: The proposed simplex kernel fails for the uniform binary case). **False displayed representation.** Source: [19], lines 130–140.

For the already included uniform $d = 2, s = 2$ model, direct finite evaluation gives $Z_0 = 1$, $Z_1 = 1/2$, and $Z_2 = 3/8$. The proposed simplex kernel is $X = q_1^2 + q_2^2 \geq 1/2$ on $q_1 + q_2 = 1$. A positive measure with mass 1 and first moment $1/2$ must have $X = 1/2$ almost surely, forcing second moment $1/4$, not $3/8$. This refutes the literal displayed kernel representation, not every possible positive moment representation.

Audit annotation 4.29 (F29: The universal initial-index negative-sign assertion fails at order 21). **Certified counterexample.** Source: [17], lines 97–117.

The source asserts $S_k(0) < 0$ for every $k \geq 6$. The review checks the single order $k = 21$ using only $a_0, \ldots, a_{21}$ from the existing data range. Exact rational logarithm intervals certify

$$0.010409191634282788066473838299027097 82 < S_{21}(0)$$

$$S_{21}(0) < 0.010409191634282788066473838299027097 84.$$

For each positive integer x, write $x = 2^b y$, $1 \leq y < 2$, and use the positive series for $\log y = 2\,\mathrm{artanh}((y-1)/(y+1))$ with an explicit rational tail bound. The reviewer stores all certificate endpoints and an independent checker. This also refutes A000011, line 199, at $U_{20}(1) = S_{21}(0)/21 > 0$.

Audit annotation 4.30 (F30: The scaled-difference table does not match the archived definition). **Numerical table errors.** Source: [12], lines 156–166.

Checks within the already archived data range give the following approximate values:

k	$4^k S_k(4)$	$100^k S_k(100)$
3	1.28628052	3.78172144
4	1.95813786	10.91995179
5	3.38854594	41.64138756
6	5.71898720	196.61369704

For example, the source instead prints $4^3 S_3(4) = 3.052$ and $4^6 S_6(4) = 97.805$. Its table also lists sixth-order values above 240 while claiming convergence from below to 240. Other ratio examples are inaccurate: A000001, line 43, prints $r_{60} \approx 26.78$ instead of 26.1132505745. The correct differences $27 - r_{60}$ and $27 - r_{100}$ are approximately 0.8867494255 and 0.5352179260, unlike A000002, lines 45–46. The exact archived integer tables themselves pass verification.

Audit annotation 4.31 (F31: The general cubic growth sketch prints the wrong exponential base). **Exponential-rate error.** Source: [18], lines 25–37.

The sketch writes $A_n^{(d)} \sim C_d d^n$ times an algebraic factor, then concludes that the ratio tends to d^3. The displayed growth rate would instead give base d when used as the asserted algebraic asymptotic, and already contradicts the verified $d = 3$ ratio limit 27. This is an error in the printed formula, not a refutation of the separate proposed ratio limit.

Audit annotation 4.32 (F32: The first weighted logarithmic formula contradicts the next line). **Sign and factor errors.** Source: [19], lines 29–35.

The expression $(1-s)\log(1/n^{d-1})\,(d-1)/2$ in line 31 equals $+(s-1)(d-1)^2\log n/2$. It has the opposite sign from line 33 and an extra factor $d-1$. For $s>1$ it is also incompatible with $Z_n(s)\le 1$, which follows from $0\le P_n(\mathbf{i})\le 1$. This literal formula is false independently of the unresolved proof status of the intended general asymptotic statement.

5. Exact checks of archived certificates

5.1. **The third-order recurrence.** For the differential equation of Φ, the original script acts exactly on $z^m\Phi^a(\theta\Phi)^b(\theta^2\Phi)^c$, where $a+b+c=3$, using the Leibniz rule. It constructs

$$R_j(n)=P_j(n)\big((n+1)\cdots(n+j)\big)^3,\qquad L=\sum_{j=0}^{3}z^{3-j}R_j(\theta-j).$$

Execution of the original script[25] leaves zero nonzero basis coefficients in $L(\Phi^3)$. Together with coefficient extraction, this finite symbolic identity proves the displayed recurrence. The proof is supplied by the identity, not by the separate check of 198 numerical residuals.

5.2. **The Newton ratio envelope.** The archived bounds are

$$L_n=27-\frac{54}{n}+\frac{48}{n^2}-\frac{20}{n^3},\qquad U_n=27-\frac{54}{n}+\frac{48}{n^2}+\frac{20}{n^3}.$$

The ratio map is

$$F(x,y;n)=-\frac{P_0(n)+P_1(n)x+P_2(n)xy}{P_3(n)xy}.$$

For $n\ge 20$, $F_x<0$, while the sign of F_y is the sign of P_0+P_1x. Substitution shows that both $-(P_0+P_1L_{n+1})$ and $P_0+P_1U_{n+1}$ have positive-coefficient numerators and denominator $(n+1)^3>0$. Thus the mixed-corner choices in the source are valid. The initial values $n=20,21,22$ pass exact rational comparisons.

For the unreduced numerators N_ℓ, N_u defined in the original script[24], the exact degrees are 12, the thirteenth differences vanish, and each initial forward difference of order zero through twelve is strictly positive. The reduced numerators have degree 9; this is consistent with the unreduced degree-twelve certificate. Some exact checks are

$$\begin{aligned}N_\ell(20)&=75340506678287596696,\\ N_u(20)&=5895157450831107829064,\\ \Delta^{12}N_\ell(20)&=4345502515200,\\ \Delta^{12}N_u(20)&=278112160972800.\end{aligned}$$

Newton's formula

$$N(20+t)=\sum_{j=0}^{12}\Delta^jN(20)\binom{t}{j}$$

has nonnegative terms and a positive constant term at every integer $t\ge 0$. This covers the entire integer tail, not a sparse sample. The positive denominators, correct corner analysis, and these certificates close the archived ratio-envelope induction.

5.3. **The archived criteria for S_2 and S_3.** The archived script[24] proposes $S_2(n) \geq \log(L_{n+2}/U_{n+1})$. The numerator of this rational difference is

$$54n^4 + 188n^3 + 90n^2 - 276n - 252.$$

After $n = 20 + t$, its descending coefficient list is

$$(54, 4508, 140970, 1956924, 10174228),$$

which is strictly positive. The denominator is $(n+1)^3(n+2)^3 > 0$. The finite segment passes the correct integer comparisons. Hence this existing envelope route confirms $S_2 > 0$ at every index.

The stage-two review[16] gives the criterion

$$H_3(x, y; n) = P_3y^3 + P_2xy + P_1x + P_0 > 0, \quad x = r_{n+1}, \quad y = r_{n+2}.$$

Within the archived ratio rectangle, the checks

$$-(P_1 + P_2L_{n+2}) > 0, \qquad 3P_3L_{n+2}^2 + P_2U_{n+1} > 0$$

establish $H_{3,x} < 0$ and $H_{3,y} > 0$. Its lower corner is therefore $H_3(U_{n+1}, L_{n+2}; n)$. After positive denominators are cleared, every numerator coefficient after $n = 20+t$ is positive. The complete coefficient lists are in the accompanying JSON; their sizes are:

Archived criterion	Degree	Coefficients
$-(P_1 + P_2L_{n+2})$	9	10
$3P_3L_{n+2}^2 + P_2U_{n+1}$	15	16
$H_3(U_{n+1}, L_{n+2})$	15	16

Together with the existing finite segment, these checks verify $S_3(n) > 0$ for every $n \geq 0$. Only the criteria already proposed in the archive were checked; no new family of certificates was searched.

5.4. **Weighted identity and local corrections.** The principal statement is

$$U_k(n) = \frac{1}{n\binom{n+k}{k}} \sum_{j=0}^{n-1} \binom{j+k}{k} S_{k+1}(j). \tag{5.1}$$

Starting with the recurrence already printed in the source, $(n+k+1)H_{n+1} = nH_n + \Delta^{k+1}f_n$, checking $V_n = n\binom{n+k}{k}H_n$ gives

$$V_{n+1} - V_n = \binom{n+k}{k}\Delta^{k+1}f_n.$$

This algebraic check identifies the integrating-factor error in the source. With $f_0 = 0$ and any finite extension of g_0, the initial nH_n term vanishes, and the telescoping sum agrees with the principal formula. The formula is therefore correct, but the erroneous intermediate expression in source line 56 is not accepted. The verified S_2, S_3 signs and (5.1) give $U_1, U_2 > 0$, including strict decrease of b_n. For larger k, the same identity does not supply the missing full-tail signs of S_{k+1}.

5.5. **Negative-square and finite counterexamples.** Direct substitution into the original data gives

$$L\big((x-1)^2(x-11)^2(x-20)^2\big) = -206, \qquad \det[a_{i+j}]_{i,j=0}^{3} = -49203288.$$

Each is an independently checkable finite obstruction to a positive real moment representation. The already proposed stage-six counterexamples give

$$2 \cdot 488 - 34^2 = -180, \qquad 1141^3 - 31^3 \cdot 52111 = -66992580.$$

They refute the relevant universal positivity statements; they do not constitute complete classifications of the parameter space.

6. Coverage and reproducibility

All current assets are covered below. Historical failures do not invalidate a later certificate that actually passes; a later final-version label does not cure an older unresolved gap. The bibliography translates the Chinese source titles for identification.

Asset	Subject	Ledger / annotations
A000001	Trinomial cubic sums: theorems and conjectures	C01
A000002	Computational benchmarks and numerical verification	C23, F24
A000003	Proofs of the bound and limits, and monotonicity conjectures	C03, C04
A000004	Third-order recurrence and monotonicity proofs, revised	C07, C08, C09, C15
A000005	Factorization analysis of P1 and P2	C10, C11, C12
A000006	Derivation of the recurrence from the differential equation (Q1), revised	C02, C06
A000007	Recurrences for k colors and m-th powers (Q3), revised	C39, C40
A000008	Mathematical analysis of the rigorous derivation in Q1	C02, C06, F24
A000009	Proofs of the ratio limit and strict ratio increase	See dependencies
A000010	Stage two: sign classification of logarithmic differences	C20, C22
A000011	Stage three: sign classification after taking roots	C24, C27, C28, C29, C30, C31
A000012	Stage two, revised: explicit thresholds and tail positivity	C21
A000013	Q1–Q3: differential derivation, factorization, and generalization	C41, C42
A000014	Stage two, revised: tail positivity with explicit errors	C17–C23, F08
A000015	Stage two, final version: Newton-series sign classification	C05, C13, C18, C19

Asset	Subject	Ledger / annotations
A000016	Stage-two independent review and attempted gap repair	C14, C16, C17, C23
A000017	Stage four: fixed-order positivity, common tails, and moments	C32, C33, C34, C35, C36, C37, C38
A000018	Stage five: d-color sums and critical positivity order	C43, C44, C45
A000019	Stage six: weighted sums, parameter classification, and counterexamples	C46, C47, C48, C49
A000020	Stage seven: a unified theory and open questions	C50, C51, C52, C53
A000021	Stage-three independent proof review	C25, C26

6.1. **Computational ranges.** The defining Franel convolution was independently evaluated through $n = 200$ and compared with both archived tables. Exact integer comparisons for $S_2, \dots, S_6$ cover $n \le 198, 197, 196, 195, 194$, respectively. A separate 120-digit Decimal scan within the already claimed range $n \le 5000$ reproduces the archived negative-index pattern. It is still classified as numerical evidence; many decimal digits are not described as a formal interval certificate.

For $U_4(4)$ and $U_5(1), \dots, U_5(4)$, the rational coefficients of $\log a_{n+j}/(n+j)$ were cleared and integer powers compared. These computations check existing assertions only. Original scripts were executed only after checking that they would not write into the research inputs; other computations were kept in the audit output directory.

6.2. **Evidence files and integrity of the inputs.** The accompanying files include `checks/exact_checks.py`, `checks/exact_results.json`, `checks/newton_certificate_results.json`, and `checks/existing_S2_S3_certificates.json`. The latter two retain the exact coefficient data for the original Newton numerators and the archived S_2, S_3 criteria. `checks/series_crosscheck.json` records a formal cross-check. A formal series truncated at c_5 is checked only through the orders it is meant to match; the next-order residual is not misidentified as a counterexample. `checks/original_source_hashes.json` records SHA256 hashes for all 2636 ordinary files in the two original export trees. A final integrity comparison is stored separately; its purpose is to establish that this audit did not alter the inputs.

The author is DeepMath06 - number theory. The English edition uses Latin Modern for text and mathematics and requires no CJK font substitution. Verified MSC codes and keywords were not supplied and are not invented. No submission, novelty, or priority claim is made.

7. Verification record and scope

Independent checks and corrections. An independent implementation recomputed the symmetric-cube operator using polynomial coefficients and Horner evaluation, obtaining zero residual. The unreduced degree-twelve Newton certificates, mixed-corner signs, archived S_2, S_3 criteria, weighted identity, finite signs, and negative-square obstruction also passed. The cross-check supported the P/L classifications

of the four original low-order conclusions and retained G for the unresolved S_4–S_6 full tails and their dependent conclusions.

Three verdicts in the frozen draft were strengthened to F:

(1) C52: the central-binomial case $(d, s) = (2, 2)$ has a common positive tail; see F27. The special cubic case C33 remains G.
(2) C49: the literal simplex kernel contradicts the uniform binary moments $1, 1/2, 3/8$; see F28. This does not rule out a different moment representation.
(3) C34: exact rational logarithm intervals certify $S_{21}(0) > 0$, also giving $U_{20}(1) > 0$; see F29. The source's finite order-six-through-twenty negatives are retained as finite evidence.

The final ledger keeps 53 assertion units: P20, L4, G14, N2, F10, O3. The source locators and tally have been checked. Additional findings are incorporated into F04 and F25–F32. The final report has 32 numbered annotations, with the same identifiers used for its PDF comment notes.

Scope of verification. The verification covered the English text and relevant TeX, all 21 current assets, and relevant candidate scripts. The separate checker and exact interval certificate are retained as `review/independent_review_checks.py` and the corresponding JSON. The primary audit independently rechecked the two rational endpoints for the targeted order-21 counterexample and obtained identical endpoints.

This is a mathematical and source audit, not formal verification, external specialist peer review, or an assessment of novelty. The additional cross-check did not rerun every historical script or the separate 5000-index decimal scan, prove the missing high-order tails, or search new recurrences or parameter regions. The original research inputs and frozen drafts remain unchanged.

References

1. Local exported research record A000001, revision 1, *Trinomial cubic sums: theorems and conjectures*. Chinese source; title translated. `records/assets/A000001.md`.
2. Local exported research record A000002, revision 1, *Computational benchmarks and numerical verification*. Chinese source; title translated. `records/assets/A000002.md`.
3. Local exported research record A000003, revision 1, *Proofs of the bound and limits, and monotonicity conjectures*. Chinese source; title translated. `records/assets/A000003.md`.
4. Local exported research record A000004, revision 3, *Third-order recurrence and monotonicity proofs, revised*. Chinese source; title translated. `records/assets/A000004.md`.
5. Local exported research record A000005, revision 2, *Factorization analysis of P1 and P2*. Chinese source; title translated. `records/assets/A000005.md`.
6. Local exported research record A000006, revision 4, *Derivation of the recurrence from the differential equation (Q1), revised*. Chinese source; title translated. `records/assets/A000006.md`.
7. Local exported research record A000007, revision 4, *Recurrences for k colors and m-th powers (Q3), revised*. Chinese source; title translated. `records/assets/A000007.md`.
8. Local exported research record A000008, revision 1, *Mathematical analysis of the rigorous derivation in Q1*. Chinese source; title translated. `records/assets/A000008.md`.
9. Local exported research record A000009, revision 1, *Proofs of the ratio limit and strict ratio increase*. Chinese source; title translated. `records/assets/A000009.md`.
10. Local exported research record A000010, revision 1, *Stage two: sign classification of logarithmic differences*. Chinese source; title translated. `records/assets/A000010.md`.
11. Local exported research record A000011, revision 1, *Stage three: sign classification after taking roots*. Chinese source; title translated. `records/assets/A000011.md`.
12. Local exported research record A000012, revision 1, *Stage two, revised: explicit thresholds and tail positivity*. Chinese source; title translated. `records/assets/A000012.md`.
13. Local exported research record A000013, revision 1, *Q1–Q3: differential derivation, factorization, and generalization*. Chinese source; title translated. `records/assets/A000013.md`.
14. Local exported research record A000014, revision 1, *Stage two, revised: tail positivity with explicit errors*. Chinese source; title translated. `records/assets/A000014.md`.
15. Local exported research record A000015, revision 1, *Stage two, final version: Newton-series sign classification*. Chinese source; title translated. `records/assets/A000015.md`.
16. Local exported research record A000016, revision 1, *Stage-two independent review and attempted gap repair*. Chinese source; title translated. `records/assets/A000016.md`.
17. Local exported research record A000017, revision 1, *Stage four: fixed-order positivity, common tails, and moments*. Chinese source; title translated. `records/assets/A000017.md`.
18. Local exported research record A000018, revision 1, *Stage five: d-color sums and critical positivity order*. Chinese source; title translated. `records/assets/A000018.md`.
19. Local exported research record A000019, revision 1, *Stage six: weighted sums, parameter classification, and counterexamples*. Chinese source; title translated. `records/assets/A000019.md`.
20. Local exported research record A000020, revision 1, *Stage seven: a unified theory and open questions*. Chinese source; title translated. `records/assets/A000020.md`.
21. Local exported research record A000021, revision 1, *Stage-three independent proof review*. Chinese source; title translated. `records/assets/A000021.md`.
22. Project note, *Recurrence and monotonicity proofs for trinomial cubic sums*. Chinese source. `notes/recurrence_and_proof.md`.
23. Project script, `polynomial_inequality_proof.py`.
24. Project scripts, `rigorous_proof_final.py` and `rigorous_tail_bounds2.py`.
25. Project script, `scripts/verify_q1_ode_to_recurrence.py`.